



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

BIOLOGY

PAPER 1 Multiple Choice

4025/1

1 hour

SPECIMEN PAPER

Additional materials:

Multiple Choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

Calculator (Optional)

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

Answer **all** questions. For each question, there are **four** possible answers, **A, B, C** and **D**. Choose the correct answer. Record your choice in **soft pencil** on the separate answer sheet provided.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

There are **forty** questions in this paper. Each correct answer will score **one** mark. Any rough working should be done on this question paper.

This specimen paper consists of 9 printed pages.

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1 The study of living organisms and their environment

- A microbiology.
- B zoology.
- C cytology.
- D ecology.

2 The diagram shows a safety symbol.



The symbol indicates

- A the possibility of electric shock.
- B exposure to a bio-hazard.
- C allow cooling before handling.
- D limit time of exposure to equipment.

3 Which one is a test for starch?

- A Benedict's test
- B Buret test
- C Emulsion test
- D Iodine test

4 Which pair of atoms are part of a water molecule?

- A hydrogen and nitrogen
- B hydrogen and oxygen
- C carbon and oxygen
- D oxygen and nitrogen

5 The movement of solute molecules down a concentration gradient is

- A active uptake.
- B osmosis.
- C plasmolysis.
- D diffusion.

6 Transmission of impulses is an adaptation of

- A white blood cells.
- B muscle cells.
- C egg cells.
- D nerve cells.

7 Which part is found in plant cells **only**?

- A cell wall
- B cell membrane
- C cytoplasm
- D nucleus

8 Which enzyme, in washing powders, removes fat stains from fabric?

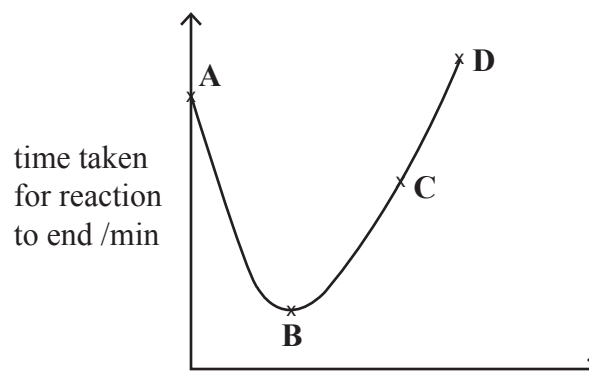
- A catalase
- B protease
- C lipase
- D amylase

9 Which one contributes to enzyme specificity?

- A temperature
- B concentration
- C shape
- D nature

10 The graph shows the effects of temperature on the souring of milk as an enzyme controlled reaction.

At which point, A, B, C or D, is the temperature optimum?



- 11 Destarching involves placing a potted plant in
- A water for 24 hours.
 - B transparent glass for 24 hours.
 - C dark cupboard for 24 hours.
 - D direct sunlight for 24 hours.
- 12 Phosphorus deficiency results in
- A stunted shoot growth.
 - B stunted root growth.
 - C brown leaf margins.
 - D premature death.
- 13 Pests affect productivity by
- A increasing pollination.
 - B disrupting soil structure.
 - C improving soil texture.
 - D reducing photosynthesis.
- 14 Cultural pests control involves the use of
- A nematodes to control numbers.
 - B wasp parasites for suppression.
 - C crop rotation to break life cycles.
 - D lace wing larvae as predators.
- 15 Which one does **not** affect the rate of photosynthesis?
- A oxygen concentration
 - B temperature
 - C light intensity
 - D carbon dioxide concentration
- 16 A toddler is fed mainly sadza and vegetables in large amounts everyday. The toddler has been feeling tired and losing weight.
- This is because the diet taken lacks
- A proteins in correct proportions.
 - B fats in correct proportions.
 - C vitamins in correct proportions.
 - D mineral ions in correct proportions.
- 17 Which one is **not** a nutrient in animal nutrition?
- A carbohydrate
 - B fibre
 - C proteins
 - D vitamins

- 18** The information shows the composition of four diets.

Which diet, **A**, **B**, **C** or **D**, is suitable for a farm labourer?

- A** potatoes, protein, fruits
- B** milk, fluids, lean protein
- C** whole grain, vegetables, low fat milk
- D** nuts, lean proteins, low fat milk

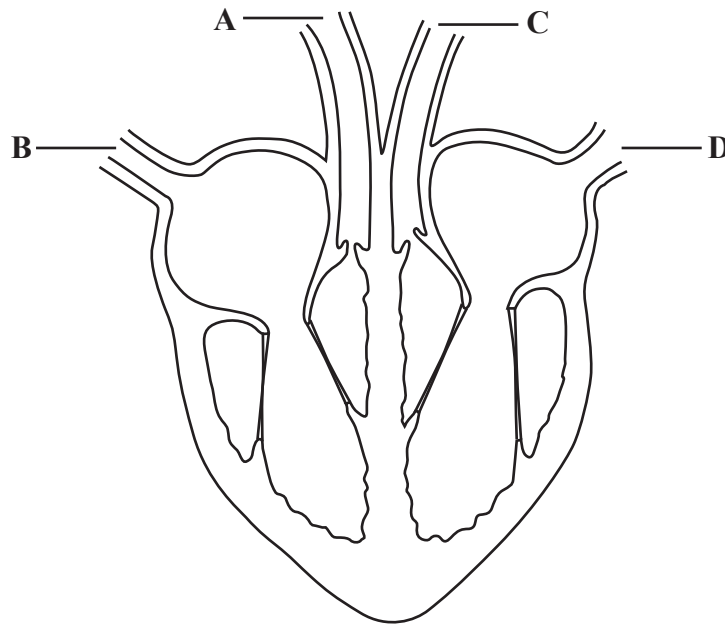
- 19** The table shows the results of food tests on four food samples.

Which food sample, **A**, **B**, **C** or **D**, contained reducing sugars?

food sample	results
A	blue-black
B	violet/purple
C	orange/brick red
D	cloudy/chalky

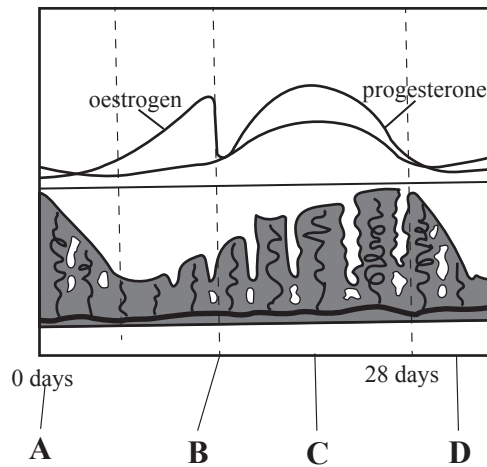
- 20** The diagram shows a mammalian heart.

Which one, **A**, **B**, **C** or **D**, is the vena cava?



- 21 The diagram shows the menstrual cycle.

Which stage, **A**, **B**, **C** or **D**, is ovulation?



- 22 Which part of the respiratory system separates the chest cavity from the abdominal cavity?

A bronchus
B bronchioles
C diaphragm
D larynx

- 23 What separates the left side from the right side of the heart?

A valves
B septum
C atrium
D ventricle

- 24 The equation shows anaerobic respiration in plants.

glucose $\longrightarrow$ carbon dioxide + **X** + energy.

What is **X**?

A water
B carbon dioxide
C alcohol
D lactic acid

- 25 Which is **not** a use of energy in living organisms?

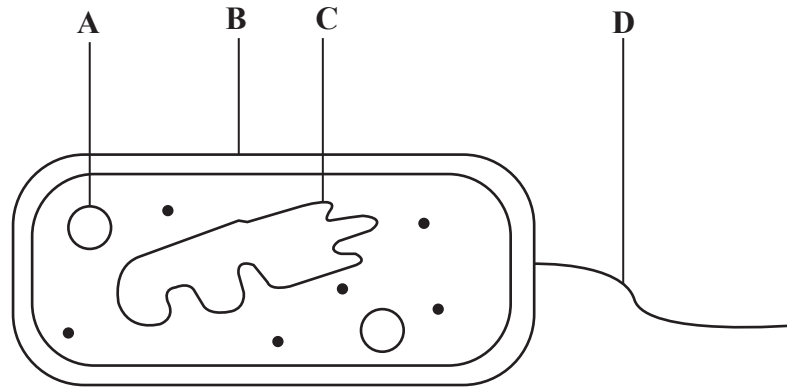
A uptake of mineral ions
B absorption of water
C conduction of impulses
D division of cells

26 Genetic material formed from more than one species is called

- A vector DNA.
- B recombinant DNA.
- C chromosomal DNA.
- D plasmid DNA.

27 The diagram shows the structure of a bacterium?

Which part, **A**, **B**, **C** or **D**, is responsible for locomotion?



28 The basic unit of inheritance is the

- A chromosome.
- B nucleus.
- C gene.
- D allele.

29 The genotype for a particular characteristic is represented by alleles **T** and **t**. **T** is dominant to **t**.

Which is correct about **Tt** and **TT**?

	Tt	TT
A	recessive phenotype	recessive phenotype
B	dominant phenotype	dominant phenotype
C	recessive phenotype	dominant phenotype
D	dominant phenotype	recessive phenotype

30 Which one is a factor of discontinuous variation?

- A temperature
- B environment
- C genes
- D nutrition

- 31 A Zimbabwean dairy farmer can improve milk production by
- A artificial selection.
 - B natural selection.
 - C sterilising cows.
 - D penning cows.
- 32 In natural selection, organisms
- A best adapted to the environment survive.
 - B that are the strongest survive and reproduce.
 - C desired by humans are allowed to reproduce.
 - D randomly mate leading to variation.
- 33 Which one is **not** a threat to biodiversity?
- A invasive species
 - B climate change
 - C subdividing land into farms
 - D creation of nature reserves
- 34 Which is a diagnostic feature of the kingdom *Protista*?
- A unicellular prokaryote
 - B multicellular eukaryote
 - C unicellular eukaryote
 - D multicellular prokaryotes
- 35 What is the source of energy in a food chain?
- A the sun
 - B carbon dioxide
 - C the water
 - D soil nutrients
- 36 Which one is a physical component of an ecosystem?
- A litter
 - B humus
 - C bacteria
 - D soil

- 37 The table shows results obtained from an experiment to determine air content of a soil sample. The volume of water used was 20 cm³.

Volume of soil only	20 cm ³
Volume of soil + water	34 cm ³

What is the percentage of air in the soil?

- A 15%
B 30%
C 59%
D 85%
- 38 A young adult develops chills, fever and sweating accompanied by shivering. The doctor suspects malaria.
- Which organism is likely causing the symptoms?
- A bacteria
B virus
C fungi
D protozoa
- 39 Which drug may cause hallucination?
- A cannabis
B nicotine
C alcohol
D paracetamol
- 40 Fungi are used in the production of
- A paracetamol.
B penicillin.
C chloroquine.
D aspirin.